# Market Analysis and Price Prediction Module for Sustainable Anthurium Farming

Project Id: 25- 26J-357

Project Proposal Report

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BSc (Hons) in Information Technology Specializing in Information Technology

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## DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

..... .....29/8/25.....

Signature of the supervisor: Date:

## ABSTRACT

Anthurium farming in Sri Lanka has also emerged as a profitable industry of floriculture, but farmers are still left with chronic issues arising from unforeseen market conditions, repeated price fluctuations, and inadequate access to reliable information. Without foresight tools, judgment overwhelmingly relies on assumptions and spur-of-the-moment information delivery, which predominantly results in insecure revenues, wasteful production planning, and foregone opportunities in local and export markets. To tackle these issues, the project proposes a Market Analysis and Price Forecasting Module that integrates innovative data-driven technologies with agricultural decision-making for the Anthurium industry.

The system will couple historical information and present-day market data with machine learning techniques to generate accurate forecasts of price fluctuations and demand trends. Methods such as time series forecasting and regression modeling will be employed to deliver real-time insights. The projections will be offered by way of a mobile application that provides farmers with uncomplicated access to projections, notifications on significant market changes, and recommendations for marketing and sales strategies. Through making advanced analytics easily understandable and actionable, the solution seeks to empower farmers to make risk-reducing and profitability-increasing proactive decisions.

The module aims to offer a structured framework for the collection and interpretation of Anthurium market information, develop flexible predictive models that learn and adapt over time as more data is incorporated, and develop an accessible platform that is flexible enough to suit practical usage by farmers with varying technical skills. Non-functional requirements such as scalability, user-friendliness, dependability, and data security are incorporated to ensure the system is efficient and sustainable. The expected result is significantly increased profitability, reduced exposure to volatile markets, and greater competitiveness of Sri Lankan Anthurium growers in the global floriculture industry.

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## LIST OF ABBREVIATIONS

- AI – Artificial Intelligence
- FMC – Firebase Cloud Messaging
- FS – Firebase Storage

# 1. INTRODUCTION

## 1.1 Background & Literature survey

### Background

Anthurium crop is a rapidly developing floriculture sector in Sri Lanka with very good prospects both domestically and internationally. However, farmers in this sector still face crucial issues that limit profitability and long-term sustainability. At the forefront of these issues are unstable market conditions, frequent price variability, and limited access to reliable market information. Without proper forecasting instruments, farmers rely on personal intuition or unofficial networks to make consequential production and sales decisions. Without organized information, this exposes them to irregular levels of income and reduced competitiveness in a continuously changing agricultural economy.

The absence of market forecasting analysis systems has made farmers vulnerable to inefficiencies such as excess output, low pricing during harvest times, and missed market opportunities. Excess yields in most cases are not consumed by the market, leading to economic loss, while shortages mean unsatisfied demand and destroyed consumer trust. Dependence on middlemen also complicates the situation since farmers cannot negotiate or plan when they take market prices. Such conditions not just lower profitability but also discourage innovation and investment in the industry, hence the growers are not in a position to attain the full economic potential of Anthurium farming.

Given these situations, the adoption of a more rational, data-driven approach has become a necessity. A market survey and price forecasting module using historical records, real-time feeds, and machine learning algorithms can provide growers with actionable information regarding price action and shifts in demand. The system would allow growers to plan their sales more effectively, reduce risks of volatility, and optimize the stability of income. With these data provided by a simple-to-use mobile app, even small farmers would enjoy advanced decision-making assistance. Ultimately, the strategy holds potential to transform Anthurium agriculture into a more lucrative and resilient venture, further catalyzing Sri Lanka's floriculture sector development.

## Literature Survey

Anthurium flower supply is highly dynamic, based on seasonally shifting demand, export levels, customers' tastes, and foreign trade dynamics. Market analysis in this case comes in handy since growers can monitor price trends, identify patterns of demand, and make appropriate production as well as selling decisions. By comparing historical and real-time market data, farmers can coordinate their cultivating cycles with market windows, optimize resource utilization, and boost profitability in a competitive ornamental flower market.

Price prediction is an essential tool for managing the financial risks associated with the Anthurium business. Time series prediction, regression analysis, and machine learning can predict alterations in market price based on historical sales, seasonal patterns, and foreign demand data. Accurate price projections allow growers to strategize harvesting, distribution, and marketing, preventing oversupply losses or market declines and maximizing overall economic performance.

Mobile applications also enhance the value of market analysis and price prediction as a function for Anthurium farmers. With mobile app provision of immediate price data, trend signals, and analytical insights on mobile phones, the applications put complex data within ready reach and ease of use. Farmers are able to respond promptly using this information, participate more effectively in local and international markets, and respond swiftly to movement in prices, achieving greater financial security and operational efficiency.

Merging market knowledge and price forecasting into a dedicated mobile system is an all-in-one solution for Anthurium growers. Such a system merges forecasting data, market data, and real-time alerts into one easy-to-use interface, which helps improve strategic decision-making, sales planning optimization, and profitability. By making data-driven intelligence accessible in real-time through the mobile system, growers are better positioned to deal with the dynamic Anthurium market and establish a presence in the flower ornamental market.

## 1.2 Research Gap

Research in agricultural forecasting has made a long way; yet, there is a substantial gap in tools particularly for the market of Anthurium flowers in Sri Lanka. Most systems available observe generalities such as e-marketing strategies, profitability, cost basis, or channels, but no system presents an integrated real-time market analysis tool tailor-made for individual farmers. Consequently, Anthurium growers lack exposure to integrative platforms that consolidate predictive price modeling, market trend following, and real-world decision-support functionalities in a single, easy-to-use platform.

Current solutions have limited ability to deliver actionable intelligence from real-time market situations. Growers find it difficult to respond to changes in export demand, seasonal cycles, and consumer trend changes, which directly affect production planning and revenues. Without a system for integration of real-time market intelligence with predictive analytics, growers cannot effectively make decisions regarding the best time for harvesting, pricing, or marketing strategies, thus constraining competitiveness and operational efficiency in the floriculture market.

The new system is also designed to fill this gap through the inclusion of a built-in price forecasting and market analysis module accessible through a mobile app. Through the incorporation of real-time market conditions, predictive analysis, and user-friendly interfaces, the system will enable Anthurium farmers to make timely informed decisions. The innovation of the proposed solution lies in its capacity to generate immediate market projections specific to the Sri Lankan Anthurium sector, allowing for strategic cultivation, sales plan, and overall profitability, thereby enhancing decision-making and local producers' competitiveness.

## 1.3 Research Problem

The Sri Lankan Anthurium flower industry is greatly hindered by the absence of real-time market analysis software for producers. Current systems do not provide timely market trend analysis, so producers cannot adequately respond to shifts in demand, seasonality, or changes in export opportunity. Lack of access to immediate data-driven information makes it difficult for growers to make decision-making choices regarding production, pricing, and sales strategy.

In addition to not being able to observe the real-time market, there is also a serious problem with predictive methods in projecting Anthurium prices. Present methods cannot combine historical information, seasonal trends, and market forces into workable price forecasts. Without proper forecasting, farmers are exposed to increased financial risk and inefficiency because they cannot anticipate market direction or optimize their cultivation and marketing strategies based on projected prices.

The combination of these two vulnerabilities—insufficiency of proper real-time market analysis and failure to anticipate the right prices—represents a significant barrier in the way of Anthurium farmers seeking to make objective strategic choices. Overcoming this problem is essential to improve operational effectiveness, reduce financial risk, and enable farmers to respond sensitively to the dynamic and competitive character of the Sri Lankan Anthurium market.

# 2. OBJECTIVES

## 2.1. Main Objective

Develop a comprehensive market analysis and price forecasting system for the Anthurium industry in Sri Lanka that benefits farmers and suppliers in making successful production, marketing, and sales decisions. The system should include historical and real-time market data, predictive analytics, decision-support functionality, and interactive AI-based tools to assist stakeholders in aligning their cultivation practices with prevailing market trends. By providing timely and accurate data on price volatility, demand seasonality, and export opportunities, the system aims to maximize profitability, minimize financial risk, and improve the competitiveness of Anthurium producers and suppliers in domestic and overseas markets. Data also has to be presented in a usable, easily interpreted format so that advanced analytics are accessible and actionable by users who have varying technical skills.

## 2.2 Specific Objectives

Gather and process historical and current market data: Compile and incorporate reliable data sets related to the Anthurium industry, including past price patterns, demand pattern seasonality, consumer trends, and export volumes. An exhaustive and recent database will serve as the foundation for sound forecasting and market analysis.

Develop forecast machine learning models: Design and train advanced machine learning models capable of forecasting market trends and predicting price movements in the Anthurium market. These models will incorporate seasonality, cycles of demand, and external market influences to generate insights that are both accurate and useful to farmers and vendors in practice.

Implement decision-making procedures: Integrate predictive knowledge in a decision-support system to guide stakeholders to prepare for cultivation, timing of harvesting, and selling strategy optimization. Such integration will allow producers and suppliers to make precautionary data-informed decisions that reduce risk, reduce loss, and ensure maximum profitability.

Develop a user-friendly mobile interface: Design a simple-to-use mobile application that allows stakeholders to access market analysis and price projections easily. The platform will utilize visualizations, including charts, graphs, and notifications, to present detailed data briefly, allowing even those without much technical know-how to understand and use the insights effectively.

Implement an AI-powered chatbot for real-time market interaction: Integrate an interactive AI-powered chatbot into the system to give instant responses to inquiries about prevailing prices, availability of vacant markets, and other vital market information. Leveraging real-time information and predictive analysis, the chatbot will enable farmers and suppliers to make timely, informed decisions, boosting responsiveness and flexibility in a dynamic market.

# 3. METHODOLOGY

## 3.1 System Architecture

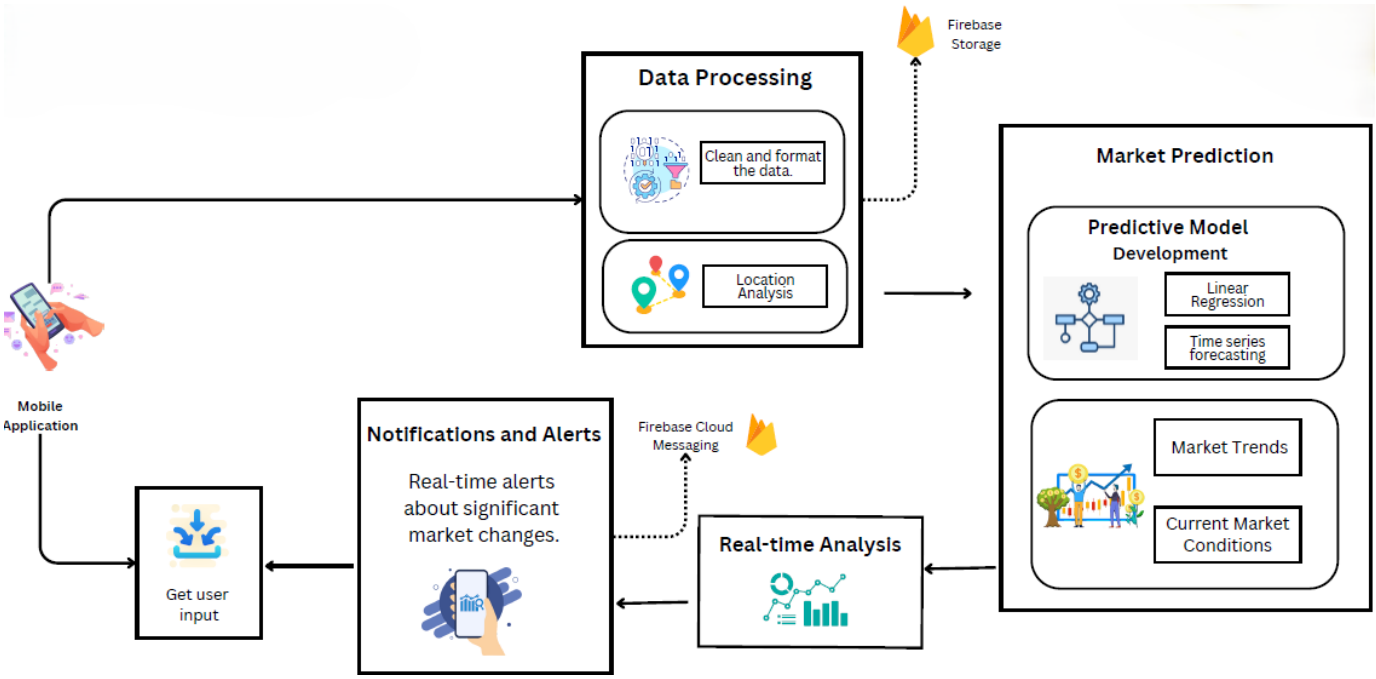


*Figure 1 -- System Architecture*

### 3.1.1 Software Solution Tools and Technologies

- Flutter
- Flask
- Google Colab
- Python
- TensorFlow
- FirebaseFirestore
- Clustering Algorithms
- Cloud Services

### 3.1.2 Commercialization

The revolutionary AI-driven Market Analysis and Price Forecast Module is designed to revolutionize decision-making across the Anthurium flower sector. Conjoining real-time marketplace data, locational insights, and advanced predictive modeling techniques, the solution empowers farmers, traders, and suppliers with accurate commodity price forecasts and market demand patterns. Sentiment analysis and domain knowledge are utilized by the platform to identify subtle market cues, identify potential opportunities, and raise alerts in a timely fashion. Apart from offering forecast accuracy, the assistant provides actionable insights that allow stakeholders to optimize sales planning, reduce financial risk, and access untapped market potential. The forward-looking ability of the technology is aligned with long-term profitability and competitiveness, paving the way for smarter and stronger agribusiness practices.

**Commercialization Strategy:**

- Strategic collaboration with farmer cooperatives, wholesale buyers, retailers, and agribusiness firms will support large-scale adoption of the system, provide access to reliable market data, and accelerate its integration into existing agricultural value chains.
- The platform will generate aggregated intelligence on commodity price movements, demand fluctuations, and sentiment trends, which can be utilized by farmers, enterprises, and policymakers as actionable decision-support data.
- Offered as a subscription-based service with tiered pricing models for individuals, cooperatives, and corporate clients, the system will be continuously optimized through user feedback, accuracy monitoring, and integration of emerging market signals to deliver increasingly precise forecasts.
- The system targets farmers, wholesalers, retailers, agribusiness companies, government agencies, and food processing firms seeking reliable market forecasts and trend analysis. Accessible through both mobile and web platforms, it provides users with real-time price updates, alerts, and forecasts from any location. Marketing will be directed toward cooperatives, production companies, and policymakers via trade fairs, expos, and conferences, while digital campaigns, targeted advertising, and partnerships with farmer associations will ensure broader outreach and trusted adoption.

#### 3.1.2.1 Future Scope

- Multi-Market Expansion: Extend the system to analyze not only local Anthurium markets but also regional and international markets, creating specialized predictive models to capture price variations across different geographies.
- Integration of External Factors: Combine sentiment analysis with climatic, logistical, and export-related data to better anticipate fluctuations in Anthurium demand and pricing, especially in global floriculture markets.
- Grower-Focused Market Education: Incorporate training resources that help Anthurium farmers interpret price forecasts, understand demand cycles, and make informed decisions on harvest timing and market selection.

# 4. PROJECT REQUIREMENTS

## 4.1 Functional requirements

- Regularly update predictive models with new data.
- Provide a mobile app interface for viewing market predictions and insights.
- Dynamic Response Adaption
- Dialog Management to provide instant responses.
- Send notifications and alerts about significant market changes.
- Enhance the user experience
- User feedback and Rating system

## 4.2 Non-functional requirements

- Performance
- Usability
- Scalability
- Security
- Compatibility
- Reliability
- Availability
- User Experience and Engagement
- Accuracy and Personalization

# 5. GANTT CHART

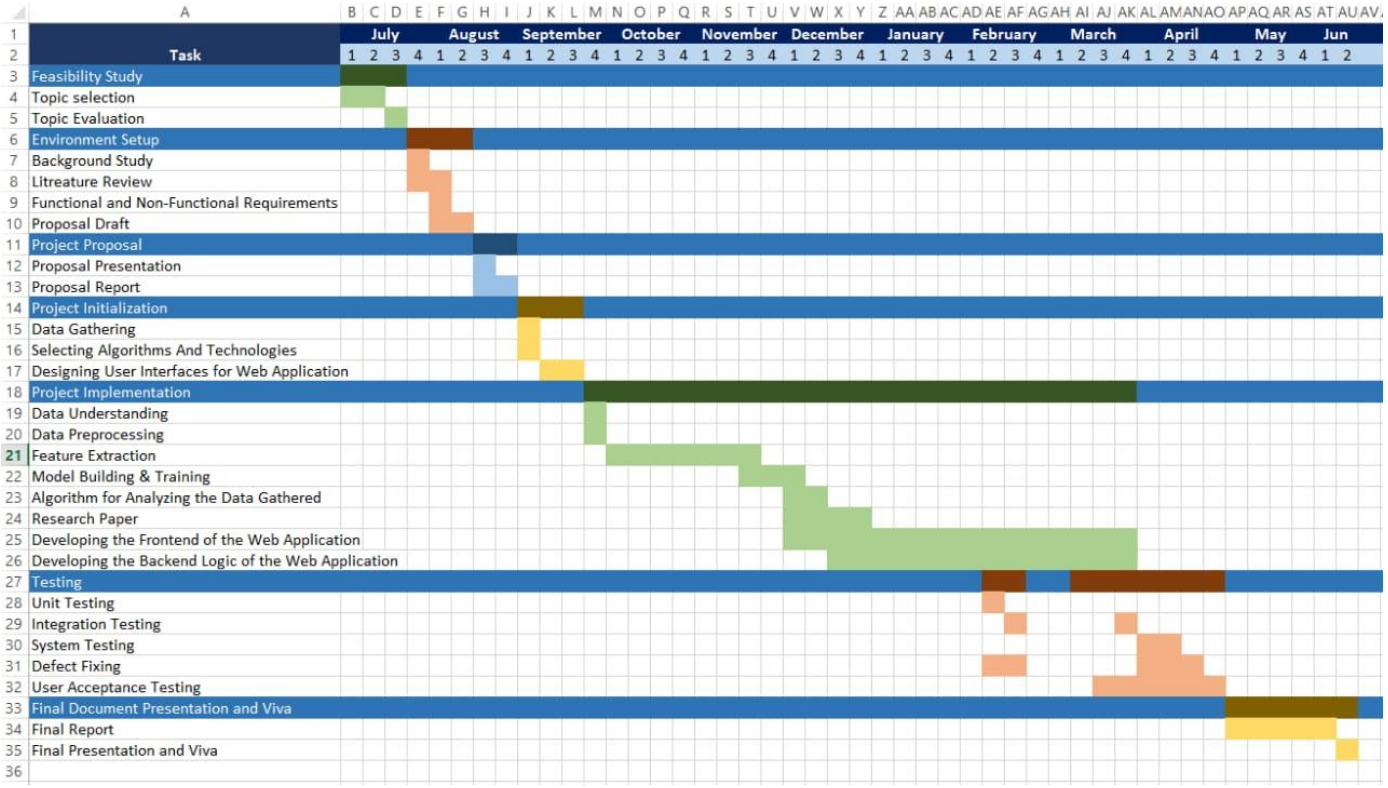


*Figure 2 -- Gannt Chart*

# 6. WORK BREAKDOWN CHART

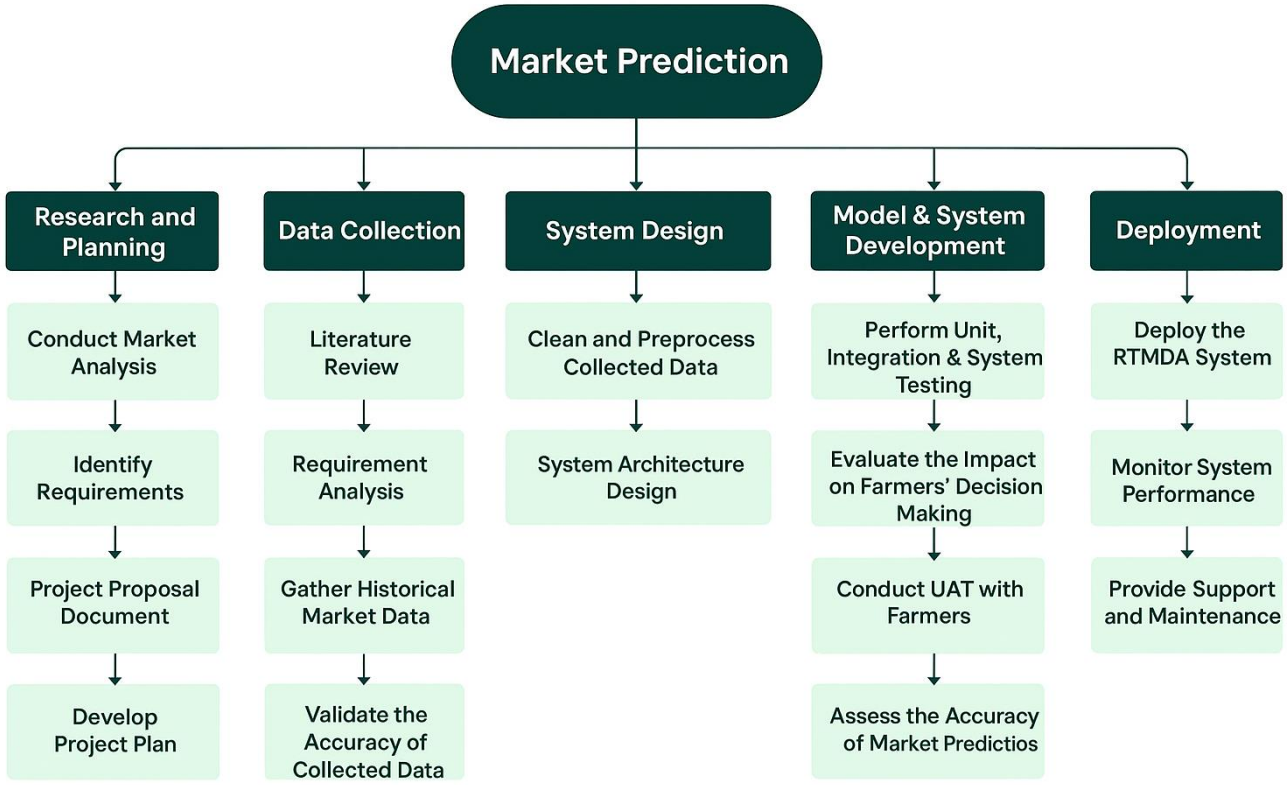


*Figure 3 -- Work breakdown Chart*

# 7. DESCRIPTION OF PERSONAL AND FACILITIES

*Table 1 --- Associated task with the component Table*

| **Sub-Component** | **Purpose** | **Associated tasks** |
|---|---|---|
| Data Collection & Preprocessing Module | Gather and preprocess historical and real-time anthurium price data from local and international markets | Collect datasets from government sources, online marketplaces, farmer associations; clean and preprocess data (remove missing values, normalize) |
| Price Prediction Engine (AI/ML Models) | Forecast future anthurium prices | Train and test models, evaluate accuracy, optimize model parameters |
| Market Analysis & Visualization System | Provide insights on demand, supply, and price fluctuations | Analyze price trends, seasonal variations, export market demand; create dashboards and graphs for decision-making |
| Decision Support & Recommendation Module | Help farmers and stakeholders make informed selling/production decisions | Generate actionable insights (best-selling time, suitable market, pricing strategy), provide alerts/reports. |
| System Testing & Maintenance | Ensure the system works efficiently and remains updated | Validate prediction accuracy, monitor system performance, update datasets and models regularly |

Facilities Required

- Laptop
- Cloud Services(FireBase, Colab)
- Basic tools for data processing

## 8. BUDGET AND BUDGET JUSTIFICATION

For this research component, we do not aim to commercialize our individual components and the whole research will be commercialized as an application. Following is the budget justification for the whole research system.

*Table 2 --- Budget Table*

| Description | Cost |
|---|---|
| Cost for Deployment | 10000.00 |
| Internet | 4000.00 |
| Travelling cost for data collection | 4000.00 |
| Google Map API | 2000.00 |
| Paper Publishing | 20000.00 |
| Total | 40000.00 |

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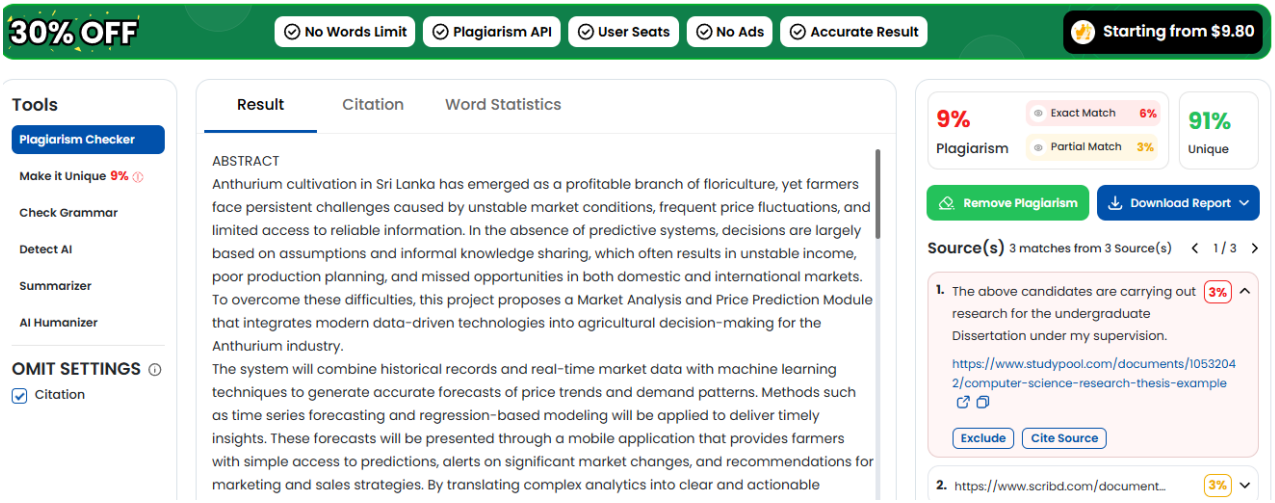
# 10. APPENDICES

## Appendix - A: Plagiarism Report




*Appendix 1 -- Plagiarism Report*